

Introduction to Blockchain and Smart Contracts

Lecture 9: Ethereum

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An Intro to “World Computer”

- Deterministic, unbounded state m/c consisting of globally accessible singleton state and a VM that applies changes to that state!
- Cryptocurrency: *Ether* (to meter and constrain execution costs)
- Vs Bitcoin:
 - Larger purpose than just a cryptocurrency: ether acts as *utility* currency
 - Programmable (Turing-complete)
 - Bitcoin tracks state of currency ownership vs Ethereum tracks the state transitions of a general purpose data store (represented as a key-value tuple)
- Birth of Ethereum: 2013 Vitalik Buterin and Gavin Wood

Components of Ethereum

- **P2P N/w**: addressable at TCP port 30303 and runs a protocol **DEVp2p**
 - DEVp2p: Does node discovery using DHT and UDP protocol; secures peer connections; allows for protocol multiplexing
- **Consensus Layer**
- **Transactions**
- **State Machine (EVM)**
- **State Storage (Google's LevelDB)**
- **Clients, Exchanges, Wallets, ...**

DEVp2p: Essentials

Node Discovery

- Launched node has **empty** peer list
 - Contacts **Bootnodes** (which are hardwired); send in a “FindNode” query
- **Distance Calculation:** Closeness is about **numerical closeness** as opposed to physical geography
 - $d(A, b) = A \oplus B$
 - Organised in to k-buckets —where each bucket stores nodes at a specific distance
- **Recursive Lookup:** Node picks a random ID or a specific target it wants to find
 - Sends “FindNode” request to k-nearest nodes it already knows
 - These nodes reply with k nodes they know that are closest to the target
 - The node takes these new nodes and repeats the process, thus getting closer to the target
- **Bonding step:** Discovered node is pinged. Once validated added permanently to the routing table

Ethereum Basics

- 1 ETH, can be subdivided further. Smallest unit is *wei*. $1 \text{ ETH} = 10^{18} \text{ wei}$
- Wallets: S/w application that manages your Ethereum account
 - Holds the keys; creates and bcasts txns on your behalf;
 - Eg:
 - **Metamask**: wallet runs in the browser (web-based)
 - **Jaxx**: multiplatform, multicurrency, mobile or desktop wallet

Getting Started with Metamask

- Navigate to <https://chrome.google.com/webstore/category/extensions>
- Search for Metamask and add it
- Click on the icon in the extensions bar and create a wallet
- Once done, Metamask creates a wallet and show you a *mnemonic backup*
 - *Consisting of 12 English words: for recovery*
- Check out your account details!
 - Address, tells which n/w you are on, ether ownership

Metamask

- You can switch n/w:
 - Main: real value, real txns, real consequence!
 - Ropsten: public test n/w, ETH has no value here
 - Kovan: uses PoA, ETH has no value, test n/w
 - Localhost 8545: Connects to a node running on the same computer as the browser.
- Use Rosten Test faucet: click on the “request 1 eth from faucet” button

Metamask

- The faucet app has created a txn. Look at the txn id.
- In a few seconds, Ropsten miners will mine your txn and your wallet will show the balance of 1 ETH.
- Click on txn id.
- The browser will take you to *Block Explorer* — a website that allows you to visualise and explore blocks, addresses and txns
- Metamask uses Etherscan block explorer
- Try sending the ETH back to the faucet.

Metamask

- Click to get info on your account> “View account in Etherscan”
 - Opens Block explorer
 - Can check the entire history of your Ethereum address
- You can also explore the txn history of other addresses
- All of this wealth of information can be explored **programmatically**

Externally Owned vs Contract Accounts

- The type of account we just created is an EOA.
- These accounts have a private key; allows control over funds and contracts
- Contract account: has smart contract code, doesn't have a private key
 - Instead it is owned and controlled by the logic of the contract code
 - Contracts have addresses just like EOAs.
 - Can send/recv ether
 - However, if the destination address of a txn is a contract account then that accounts contract gets executed on EVM

A Smart Contract

- Immutable list of rules as a computer program
- Deterministic - outcome is the same for everyone who runs it with a given Ethereum state
- **Context:** Local state, context of the txn that called it, information on recent blocks
- **Runtime:** Decentralised computer

A Simple Contract

- In Solidity
- It is also a flawed contract
- Similar to Java or C++
- Explanation of the contract!